

PHAM DUC THINH

Technical Artist Portfolio – Unreal Engine

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This document summarizes selected projects with a written breakdown of my specific role, responsibilities and technical contributions. Videos and demos are available via the portfolio link above.

Variety of Mobile Games (2012 - 2018)

Roles: Solo Game Developer, Technical Artist

Technologies / Platforms: OpenGL, XNA, libGDX, Unity – Desktop, Mobile, Web

Overview: Develop numerous cross-platform games and publish them on marketplaces. Implement extreme optimization techniques to support low-end devices.

My Contributions:

- Design and program small-scale games targeting web and mobile devices
- Create and texture, rig, animate low-poly 3D character and vehicle models
- Write custom shaders optimized for specific purposes, including skinning and morphing on demand, utilizing extreme lighting optimization (shadow map, vertex color, billboard, etc.)
- Write custom shaders for post-processing effects (blur, noise, glitch, etc.)

Urban Traffic System (2019)

Roles: Solo Game Developer, Technical Artist

Technologies / Platforms: Unreal Engine 4 – Desktop, Mobile

Project Overview: An open-world, city-based prototype in the spirit of GTA-style games. This is my self-funded project.

My Contributions:

- Develop a traffic simulation system for open-world, city-based games, including population, autonomous vehicles, mini-map and navigation features
- Standardize the united chassis (skeleton), materials, animation blueprint for vehicles
- Optimize assets, materials, and open-world city scenes, targeting low-end devices
- Edit and optimize visual effects using Cascade Particle System
- Design and develop game UI and HUD using Unreal Motion Graphics

F1 Singapore GP (2020)

Roles: Senior Developer, Technical Artist

Technologies / Platforms: Unreal Engine 4 – Desktop, Mobile

Client: Singapore Tourism Board

Project Overview: This project was created so that F1 fans could experience the Grand Prix before the race starts. It includes a complete track map to explore, and weekly competitive game events.

My Contributions:

- Continuously receive and implement ideas from director and client
- Import virtual environments received from artist into Unreal Engine, standardize assets and materials, optimize rendering performance for cross-platform support, minimize environmental collisions
- Edit and optimize particle effects using Cascade Particle System
- Develop interactive events, implement mini-map system, program weekly online mini-games

As these responsibilities are standard for Senior Developer and Technical Artist roles, I will omit them from projects with the same roles below.

Certis Cisco Metaverse for Presentation (2020)

Roles: Game Developer

Technologies / Platforms: Unreal Engine 4 – Desktop

Client: Certis Cisco

Project Overview: A metaverse application that allows B2B parties to join or host a meeting/presentation session. It includes a 3D virtual environment and a chat system.

My Contributions:

- *Responsibilities are the same as the F1 Singapore GP project above*
- Develop interactive events, integrate a video broadcast module into the application

Christmas Wonderland (2020)

Roles: Game Developer

Technologies / Platforms: Unreal Engine 4 – Desktop, Mobile

Client: BlueSky Events

Project Overview: This project was built during the COVID lockdown period and features a metaverse. Users can explore a variety of activities like carnival games, multiplayer games, watch videos, etc.

My Contributions:

- Program elves and interactive deer to move around in their enclosure, implement procedural animation techniques in the animals' movements

Once a Pirate Show (2021)

Roles: AR Game Developer

Technologies / Platforms: Unreal Engine 4 – Mobile AR

Client: Resorts World Sentosa

Project Overview: This project features 3D zones of interactive and multimedia experiences incorporated with real time 3D characters that tracks and responds to visitors' movements, simulated tavern fight scene and mobile application.

My Contributions:

- Develop and optimize Kraken application (mobile AR)
- Develop and optimize Stab the Pirate game (mobile AR)

Sandbox Exhibition (2021)

Roles: Senior Developer, Technical Artist

Technologies / Platforms: Unreal Engine 4 – Desktop

Client: Singapore Discovery Centre

Project Overview: Includes full-fledged multiplayer 3D games, particle tracking interactives, gigantic monopoly styled digital board game, AR applications, object recognition exhibits, first person real time 3D virtual simulation, etc.

My Contributions:

- *Responsibilities are the same as the F1 Singapore GP project above*
- Develop the Crisis Pod interactive game
- Develop a runtime procedural city, including buildings, foliage, for use in the SDCity application
- Develop a 3D avatar system with editable characters, and outfit interchangeable features

Vinpearl Digital Aquarium (2022)

Roles: Senior Developer, Technical Artist, On-site Technical Specialist

Technologies: Unreal Engine 4

Client: Vinpearl Phu Quoc

Project Overview: This museum features high tech features like audio guided tour, camera tracking interactives, touch wall games and our DCI Original Concept 'Aquaman Photo Opp' experiences which fishes gather behind you for a photo opportunity.

My Contributions:

- *Responsibilities are the same as the F1 Singapore GP project above*
- Using particles, materials, and post-processing techniques to depict the underwater environment
- Program interactive sharks and schools of small fishes, integrate a 360 projection mapping

Online Board Games (2023)

Roles: Senior Game Developer, Technical Artist

Technologies / Platforms: Unreal Engine 5 – Desktop, Mobile

Project Overview: A DCI's in-house multiplayer board games, created for showcasing purpose.

My Contributions:

- Develop a board game framework with standardized multiplayer techniques (replicates, RPCs)
- Develop online features (account management, energy, items, matchmaking, etc.)

- Using MetaHumans technology to create realistic models for game characters
- Implement procedural animation techniques to make characters interact with the environment naturally (picking up chess, holding cards, rolling dice, etc.)
- Optimize rendering for cross-platform support, especially mobiles
- Design game UI and HUD using Unreal Motion Graphics
- Build a simulated PlayFab matchmaking feature for testing on local
- Write automated client/server build and CDN/PlayFab deployment pipelines

Home Team Festival (2023)

Roles: Senior Developer, Technical Artist, On-site Technical Specialist

Technologies / Platforms: Unreal Engine 5 – Desktop

Client: Home Team Academy

Project Overview: Interactive experiences for the Home Team Festival 2023. It includes holographic avatar guide, VR scenario game, cycling game, generic touch interactive, quiz and trivia games.

My Contributions:

- *Responsibilities are the same as the F1 Singapore GP project above*
- Build and program custom game controllers for the Fire Fighter game
- Program and optimize the Prison Fight VR experience
- Deploy the project on-site and provide technical support throughout the event

Madness Arena (2024)

Roles: Senior Game Developer, Technical Artist

Technologies / Platforms: Unreal Engine 5 – Desktop

Project Overview: Another DCI's in-house multiplayer online battle arena game, created for showcasing purpose.

My Contributions:

- Develop online features (account management, matchmaking, etc.)
- Rig, animate, program one of the main characters, Gunner

Simple Combat Engine (2025)

Roles: Solo Game Developer, Technical Artist

Technologies / Platforms: Unreal Engine 5 – Desktop, Mobile

Project Overview: A high-fidelity flight-simulation engine intended for release on developer marketplaces. This is my self-funded project.

My Contributions:

- Develop comprehensive aircraft operations (taxi, take-off, landing) and aerodynamic flight mechanics (angle of attack, post-stall maneuvers)
- Implement air-to-air and air-to-ground combat systems
- Rig, animate, implement procedural animation techniques on aircraft

- Edit Niagara and materials to create visual effects for aircraft (jet thruster, wing viper, contrails, missile trail, bullet impact, etc.)
- Program the specialized HUD systems (flight instruments, radar, etc.)